

Final Exam (First attempt) of Academic year 2025-2026 (Copy 1)

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Notice – (1) The questions are on two pages. (2) Ten marks for each question.
(3) Answers must be written in pencil only and answer for Five Question only.

Q1- Select true answer for Ten only from the following sentence:

1. If $A = \{1, 2\}$ and $B = \{x, y\}$ then $A \times B$

- a) $\{(1, x), (1, y), (2, x), (2, y)\}$
- b) $\{(1, 2), (x, y)\}$
- c) $\{1, 2, x, y\}$

2. The number of elements in $\{1, 2, 3, 4, 5, \emptyset\}$...

- a) 4
- b) 5
- c) 6
- d) 3

3. If $A \subseteq B$, then,

- a) $A = B$
- b) All elements of A are contained within B
- c) All elements of B are contained within A
- d) No relation

4. If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$, then $A \setminus B$

- a) $\{3\}$
- b) $\{1, 2\}$
- c) $\{4\}$
- d) $\emptyset$

5. If $U = \{1, 2, 3, 4\}$ and $A = \{1, 2\}$, then A^c

-
- a) $\{1, 2\}$
- b) $\{3, 4\}$
- c) $\{2, 3\}$
- d) $\emptyset$

6. $(A \cup B) \cap A$

- a) A
- b) B
- c) $A \cap B$
- d) $\emptyset$

7. $A \cup A$

- a) $\emptyset$
- b) A
- c) B
- d) U

8. $(A^c)^c$

- a) $\emptyset$
- b) U
- c) A
- d) B

9. Which of the following is always true?

- a) $A \cup A = \emptyset$
- b) $A \cap A = A$
- c) $A \cup A = B$
- d) $A \cap A = \emptyset$

10. If $A \cup B = A$, then,

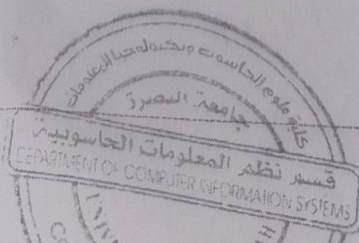
- a) $B \subseteq A$
- b) $A \subseteq B$
- c) $A = B$
- d) $B = \emptyset$

11. $(A \cap B)^c$

- a) $A^c \cap B^c$
- b) $A^c \cup B^c$
- c) $A \cup B$
- d) $A \cap B$

12. De Morgan's Law.....

- a) $(A \cup B)^c = A^c \cap B^c$
- b) $(A \cap B)^c = A^c \cap B^c$
- c) $A \cup B = A^c \cap B^c$
- d) None



3 "I just saw my old friend, and either he's grown or I've shrunk"

This meaning is:

- a. $F \wedge (Q \vee S)$
- b. $(F \wedge Q) \vee S$
- c. $F \wedge Q \vee S$

4 When is the Boolean expression $(\neg P) \wedge (\neg Q)$ TRUE?

- a. When P is false and Q is true
- b. When P and Q are true
- c. When P and Q are false
- d. When P is true and Q is false

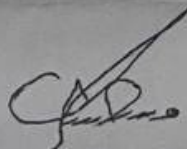
5 Which of the given options is equivalent to the following of Boolean expression?

- a. $\neg P \wedge Q$
- b. $(P \wedge Q) \wedge \neg(\neg P \wedge Q)$
- c. $P \Leftrightarrow Q$
- d. $\neg(P \rightarrow Q)$

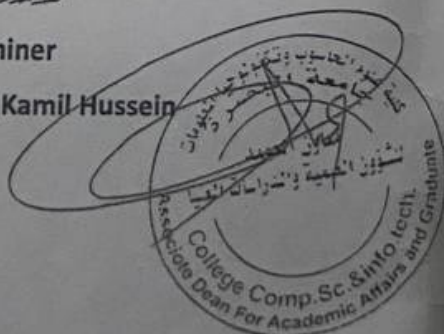
6 $GCD(406, 555) =$

- a. 1
- b. 46
- c. 23
- d. 20

END OF TEST

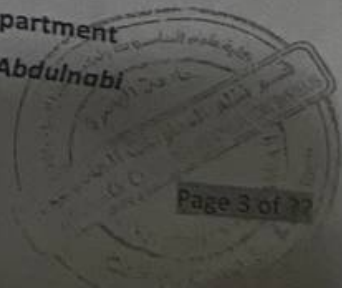

Examiner

Marwah Kamil Hussein




Head of the Department

Dr. Heidar M. Abdulnabi



Q2- A/ let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{3, 4, 7, 8, 9\}$, Find $((A \cap B) - (A \cup B))^c$, with draw

B/ Prove that using Truth Table only:

- 1- $P \rightarrow Q \equiv \neg P \vee Q$
- 2- $\neg(P \rightarrow Q) \equiv P \wedge \neg Q$

$(P \rightarrow Q) \rightarrow J$

$\frac{20}{12}$

Q3- If P represents the statement "The study is serious" and Q represents the statement "Success is guaranteed," write the following statements in logical notation:

- 1- If the study is serious, then success is guaranteed. $P \rightarrow Q$
- 2- Success is guaranteed if and only if the study is serious. $P \leftrightarrow Q$
- 3- The study cannot be serious and success is not guaranteed. $\neg(P \wedge Q)$

|||||

Q4- Use the greatest common divisor method to find:

- 1- GCD (124, 56).
- 2- GCD (406, 555).

Show the steps of the solution in detail.

Q5- If you have p and q variables, Calculate:

$$P \oplus Q, P \rightarrow Q, P \leftrightarrow Q$$

Q6- A/ Let the set $A = \{1, 2, 3\}$. If we have the following relations:

$$R = \{(1, 1), (1, 2), (2, 1)\}$$

$$S = \{(2, 2), (2, 3), (3, 2)\}$$

find the relation $Q = R \cup S$, and then determine whether Q is a symmetric relation.

B/ Let the set $A = \{1, 2, 3, 4\}$.

Let the relations R and S be defined from A to A as follows:

$R = \{(x, y) \mid x < y\}$, $S = \{(x, y) \mid x + y = 5\}$, Find the following:

- 1- $R \cap S$
- 2- $R - S$

P	Q
T	T
T	F
F	T
F	F

With best wishes for success

Head of Department

Quiz (1): let $A = \{1, 2, 3, 4, 5, 6\}$

$B = \{3, 4\},$

Find $((A \cap B) - (A \cup B))^c$, with draw

ture."Then, explain when this statement is false.

variables, Calculate:

Q3: Prove:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Q3: Prove the validity of the following statements for THREE only.

(15 M)

- 1- $A \subseteq B \Leftrightarrow A \cup B = B$
- 2- $X \wedge (Y \vee Z) \equiv (X \wedge Y) \vee (X \wedge Z)$
- 3- $(P \wedge Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \equiv (\neg P \vee Q)$
- 4- Let $A \subseteq B$ and $B \subseteq D$, then $A \subseteq D$

Q4: Let R, Q are relationships defined as follows:

(5 M)

$$R = \{(x, y) \mid (x, y) \in \mathbb{N}^* \mathbb{N}, x + 2y = 3\}$$

$$Q = \{(x, y) \mid (x, y) \in \mathbb{N}^* \mathbb{N}, 2x + 1 = 5\}$$

- 1- Dom
- 2- Codom
- 3- $R \cap Q$
- 4- R^{-1}
- 5- Rang

Q5: Choose the correct answer for any FIVE only questions.

(15 M)

1 The negation of $\neg S \vee (\neg R \wedge S)$ is:

- a. $S \wedge R$
- b. $S \wedge (R \wedge \neg S)$
- c. $S \vee (R \vee \neg S)$
- d. $S \wedge \neg R$

2 (If "I will have chicken for lunch", and "I will have steak for dinner", so, I don't Care about hunger"), That meaning is :

- a. $P \wedge Q \rightarrow \neg R$
- b. $P \vee Q \rightarrow R$
- c. $P \wedge Q \Leftrightarrow \neg R$
- d. $P \vee Q \Leftrightarrow R$



Q1: Analyze logically using the OR operator statement:

P: "The student is successful."

Q: "The student is diligent."

R: "The student attended the lecture." Then, explain when this statement is false.

Q2: If you have p and q variables, Calculate:

$$P \wedge Q$$

$$P \vee Q$$

$$P \oplus Q$$

$$P \rightarrow Q$$

$$P \leftrightarrow Q$$

Q3: Prove:

$$P \rightarrow Q \equiv \neg P \vee Q$$



Final Exam of Discrete Structure
University of Basrah
College of Computer Science and Information Technology
Department of Computer Information Systems
2024-2025/ Second Attempt



Exam Duration: 2 Hours

Date: /9/2025

- N.B.
- (1) The maximum marks for this paper is 50 [10+5+15+5+15].
 - (2) The marks for each question are shown in the right side of the question.
 - (3) Answers to questions should be grouped and written together, i.e. all answers to sub questions of individual questions like Q1, Q2, Q3 etc. should be answered one below other.
 - (4) Please read each question carefully before you start to answer it.

Q1: Answer any FIVE question:

(10M)

Let $A=\{2,3,a\}$, $R=\{(1,1),(1,2),(2,2),(3,3),(a,a),(2,1)\}$

Is R an equivalence relationship? Explain that.

Let $A=B$, and $A=\{a \mid a-2a+3=0\}$, Find $A \cap B$

$A=\{2,3,5\}$, $B=\{3,5,8,9\}$

Let $R: A \rightarrow B$, $a \in A$, $b \in B$,

if $(a \bmod b=0)$, find R^*

$U=\{x \mid x \text{ is alphabet English}\}$,

$A=\{a,b,c\}$, $B=\{a,i,o,u,e\}$, find A^c , B^c

GCD (450,60), by first way.

$\neg R \vee P \vee Q$

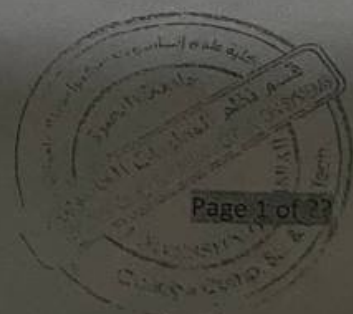
Q2: Find answer the following of question:

(5 M)

$A=\{1,2\}$, $B=\{a,b\}$, $C=\{0,1\}$

$A \cap A^c = \dots\dots\dots$

$(A \cup B)^c = \dots\dots\dots$



Q1: \\\\ Let $A = \{1, 2, 3\}$, $B = \{b \mid 10 - 2b = 2\}$, R is half relationship from A to B
Find $A \cup B$, R^* , R^{-1} , with draw for each result.

Q2: \\\\ Let R, Q are relationships defined as follows:

$$R = \{(x, y) \mid (x, y) \in \mathbb{N}^* \mathbb{N}, x + 2y = 3\}$$

$$Q = \{(x, y) \mid (x, y) \in \mathbb{N}^* \mathbb{N}, 2x + 1 = 5\}$$

Find: 1- $R \cap Q$

2- R^{-1}

3- Rang

Q3: \\\\ Let $R = \{(1, 2), (2, 3)\}$, $S = \{(2, 1), (3, 2)\}$, $Q = R \cup S$
o'vline
Is Q : Symmetrical relationship? *o'vline* Transitive relationship? Then find Q^{-1}



- N.B. (1) The maximum marks for this paper is 50 [10+10+15+15].
(2) The marks for each question are shown in the right side of the question.
(3) Answers to questions should be grouped and written together, i.e. all answers to sub questions of individual questions like Q1, Q2, Q3 etc. should be answered one below other.
(4) Please read each question carefully before you start to answer it.

Q1: Choose the correct answer for any FIVE only questions.

10
[2x5]

1- When is the Boolean expression $(\neg P) \wedge (\neg Q)$ TRUE?

- a. When P is false and Q is true c. When P and Q are true
b. When P and Q are false d. When P is true and Q is false

2- The negation of $\neg S \vee (\neg R \wedge S)$ is:

- a. $S \wedge R$ b. $S \wedge (R \wedge \neg S)$ c. $S \vee (R \vee \neg S)$ d. $S \wedge \neg R$

3- Which of the given options is equivalent to the following of Boolean expression?

- a. $\neg P \wedge Q$ b. $(P \wedge Q) \wedge \neg(P \wedge Q)$ c. $P \leftrightarrow Q$ d. $\neg(P \rightarrow Q)$

4- (If "I will have chicken for lunch", and "I will have steak for dinner", so, I don't Care about hunger"), That meaning is :

- a. $P \wedge Q \rightarrow \neg R$ b. $P \vee Q \rightarrow R$ c. $P \wedge Q \leftrightarrow \neg R$ d. $P \vee Q \leftrightarrow R$

5- One of the following operands is called Unary operand:

- a. Conjunction operator c. Implication operator
b. Negative operator d. Exclusive operator

6- "I just saw my old friend, and either he's grown or I've shrunk"

This meaning is:

- a. $F \wedge (Q \vee S)$ b. $(F \wedge Q) \vee S$ c. $F \wedge Q \vee S$

7- $GCD(406, 555) =$

- a. 1 b. 46 c. 23 d. 20

Q2: Answer any FIVE questions.

15
[3x5]

- 1- $\neg R \vee P \vee Q$
- 2- Let $A=B$, and $A=\{a \mid a-2a+3=0\}$, Find $A \cap B$
- 3- GCD (450,60), by first way.
- 4- $U=\{x \mid x \text{ is alphabet English}\}$, $A=\{a,b,c\}$, $B=\{a,l,o,u,e\}$, find A^c , B^c
- 5- $A=\{2,3,5\}$, $B=\{3,5,8,9\}$, Let $R: A \rightarrow B$, $a \in A$, $b \in B$, if $(a \bmod b=0)$, find R^*
- 6- Let $A=\{2,3,a\}$, $R=\{(1,1),(1,2),(2,2),(3,3),(a,a),(2,1)\}$
Is R an equivalence relationship? Explain that.

Q3:

A: Briefly type of relationship for each of the following sets, any FIVE terms.

5
[1x5]

1. = $\{a,b,c \in A, (a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R\}$ ✓
2. = $\{(b,a) \mid a \in A, b \in B \wedge (a,b) \in R\}$ ✓
3. = $\{x \mid x \neq x, x \text{ is number}\}$ ✓
4. $A=\{1,2\}$, $B=\{a,b\}$, $C=\{0,1\}$
..... = $\{A,B,C\}$ ✓
5. = $\{x \mid x \in A \vee x \in B\}$ ✓
6. $A \cap A^c = \dots\dots\dots$
7. $(A \cup B)^c = \dots\dots\dots$

B: Let R, Q are relationships defined as follows:

5
[1x5]

$$R=\{(x,y) \mid (x,y) \in N^*N, x+y=5\}, Q=\{(x,y) \mid (x,y) \in N^*N, 2x-y=4\}$$

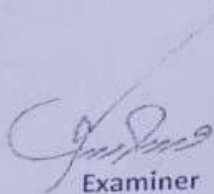
Find : 1- Dom 2- Codom 3- $R \cap Q$ 4- R^{-1} 5- Rang

Q4: Prove the validity of the following statements for THREE only.

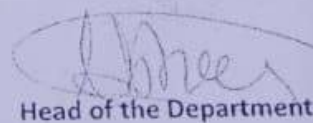
15
[5x3]

- 1- $A \subseteq B \Leftrightarrow A \cup B = B$
- 2- $X \wedge (Y \vee Z) \equiv (X \wedge Y) \vee (X \wedge Z)$
- 3- $(P \wedge Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \equiv (\neg P \vee Q)$
- 4- Let $A \subseteq B$ and $B \subseteq D$, then $A \subseteq D$

END OF TEST


Examiner

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